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Optimization Algorithms WS2526

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1 Optimization Algorithms – WS2526

Programming Exercise

Placement Strategy: Bottom–Left The placement algorithm uses the classical Bottom–Left heuristic to position rectangles inside boxes.

Ordering Strategy (Greedy)

- **Area Descending**
  - If equal area, the order is refined by descending side length.

Neighborhood (Local Search) The local search explores multiple neighborhood structures:

- **Geometry:** unpack half of the boxes and repack them.
- **Permutation:**
  - Swap two random rectangles.
  - Reverse a random subsequence.
- **Overlap:**
  - Resolve overlaps.
  - Repack most–overlapping box.
  - Repack least–used box.
  - Repack most–used box.
  - Move largest rectangle out of most–overlapping box.

2 Tuning the Algorithms and Neighborhoods

The pure versions of the algorithms, as presented in the lecture, are too slow when the entire neighborhood is evaluated in every iteration. Following the assignment guidelines, we applied two main tuning strategies.

Early Stopping

To avoid wasting time in stagnating regions, the local search stops early whenever no improvement is found for **10 consecutive iterations**. Whenever an improvement occurs, the counter resets. This simple rule significantly reduces runtime with only a minor effect on solution quality.

Reducing the Neighborhood

Instead of evaluating all neighbors, only a small, heuristically selected subset is considered. Priority is given to moves involving:

- boxes with overlaps,
- boxes with high occupancy,
- or randomly chosen diverse candidates.

This follows the suggestion that the neighborhood does not need to be explored fully, but only in the most promising directions. The result is a much faster local search while maintaining competitive solution quality.

3 Results – Full Comparison of All Methods

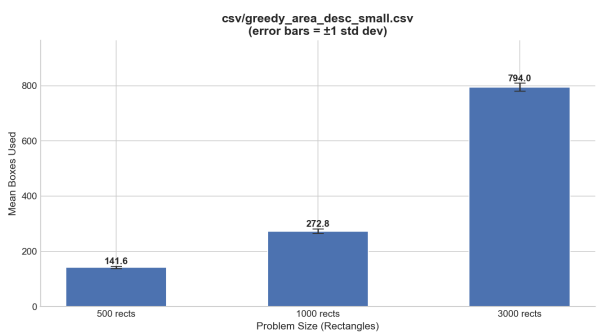
This section compares all placement and improvement strategies:

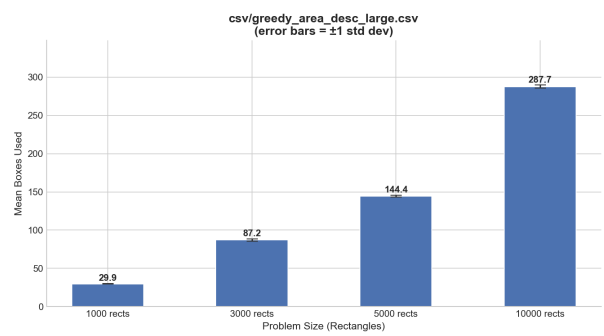
- Greedy (Area Descending, Side Descending)
- Local Search (Geometry, Overlap, Permutation)
- Small vs Large instances

Each metric is shown in its own section. Within each metric, all algorithms are compared in a consistent 2-per-row layout.

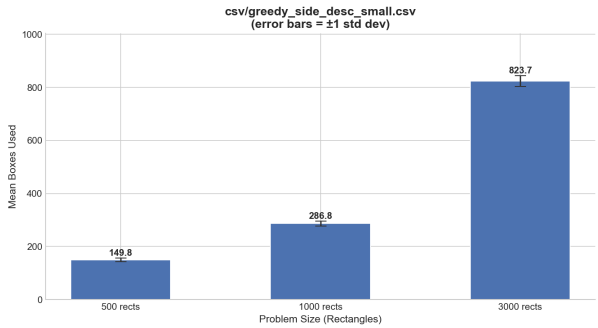
4 Mean Boxes Used

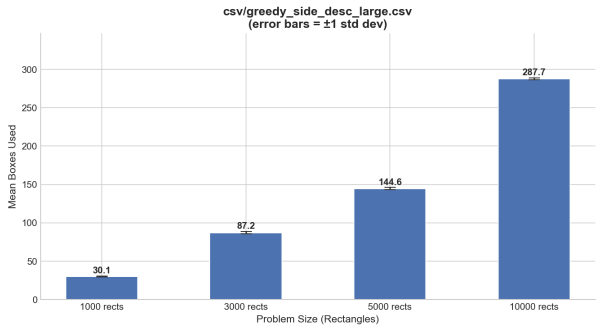
Greedy – Area Descending



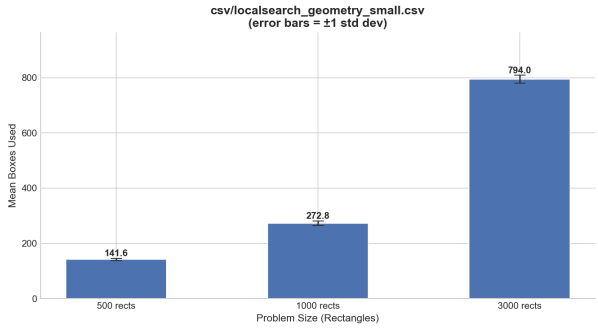


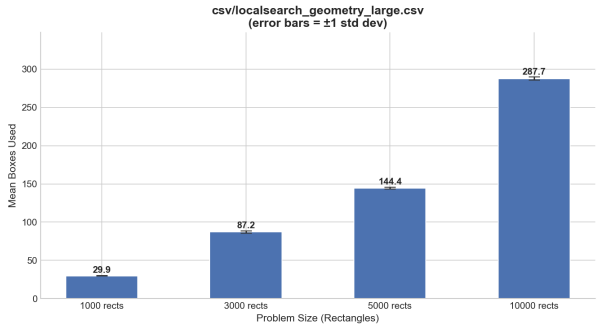
Greedy – Side Length Descending



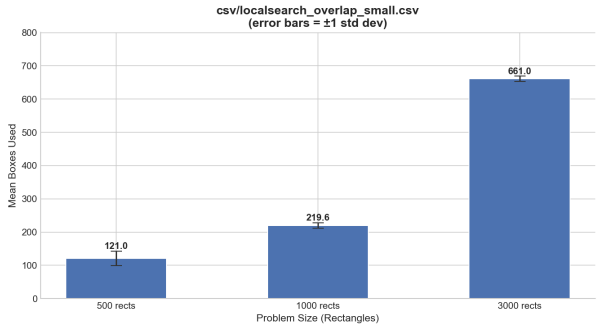


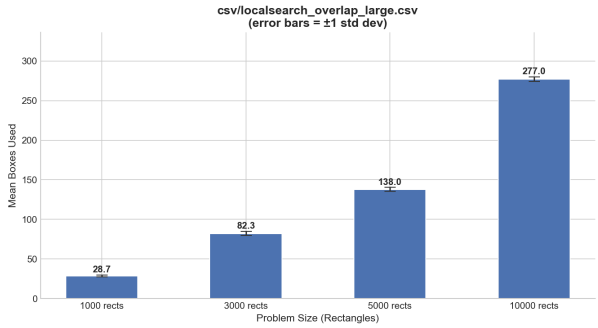
Local Search – Geometry



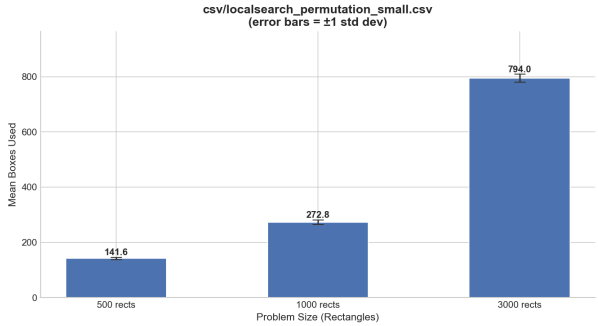


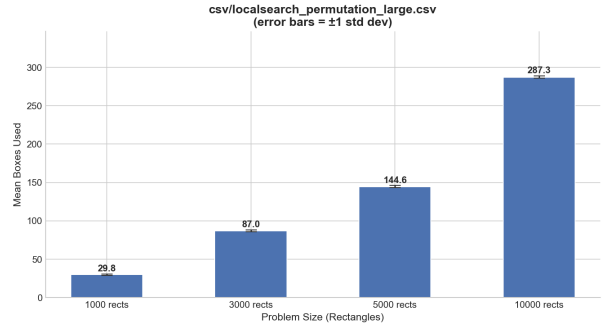
Local Search – Overlap





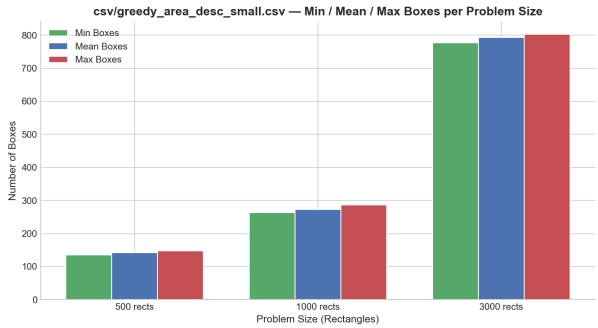
Local Search – Permutation

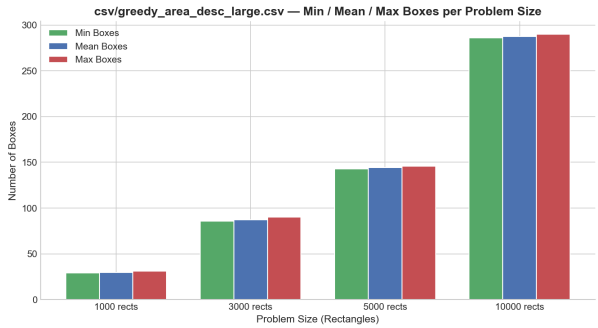




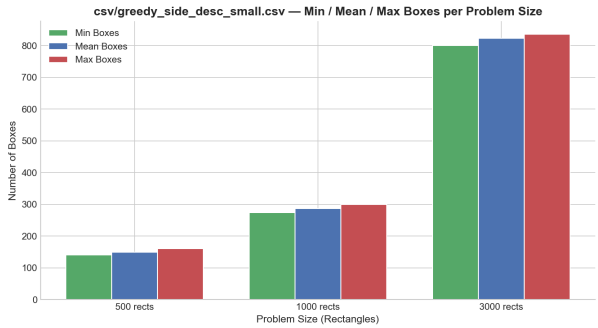
5 Min / Mean / Max Boxes

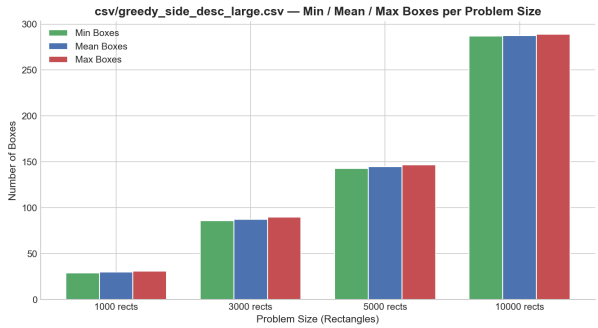
Greedy – Area Descending



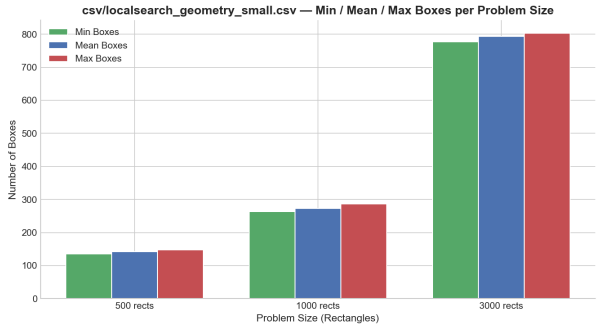


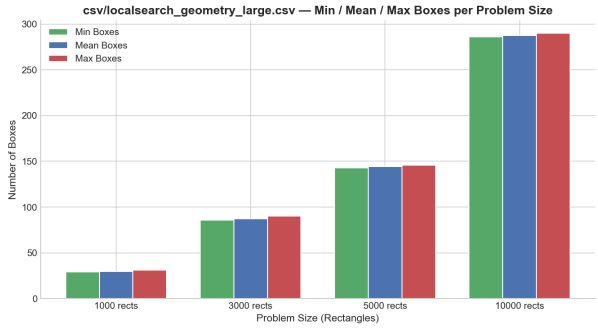
Greedy – Side Length Descending



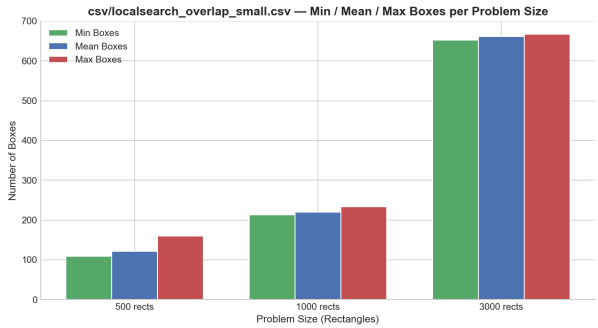


Local Search – Geometry

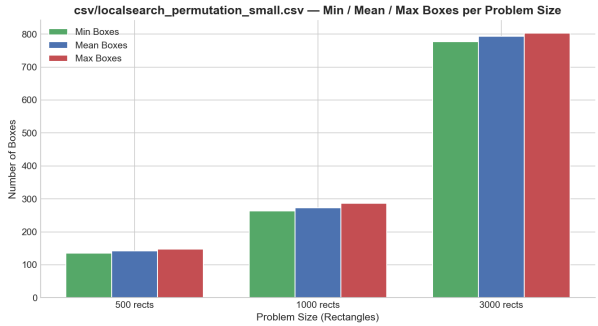


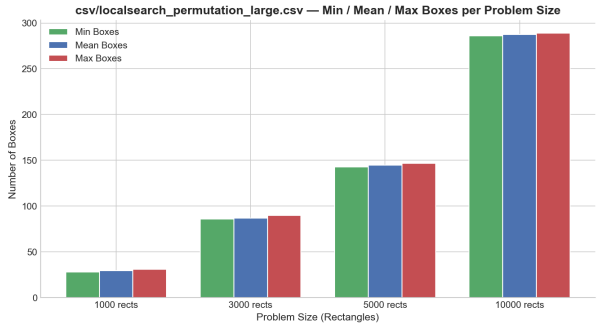


Local Search – Overlap



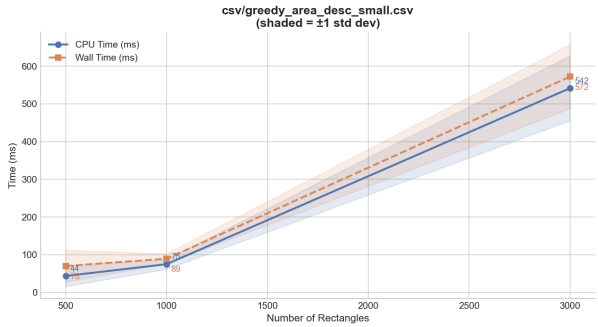
Local Search – Permutation

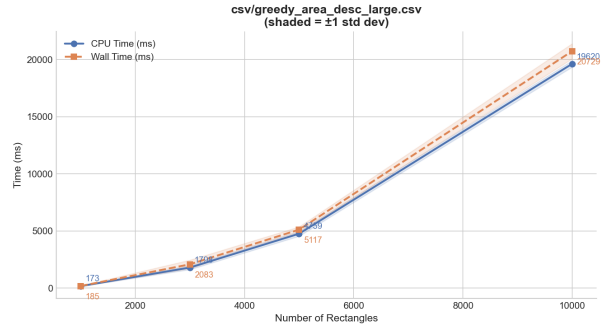




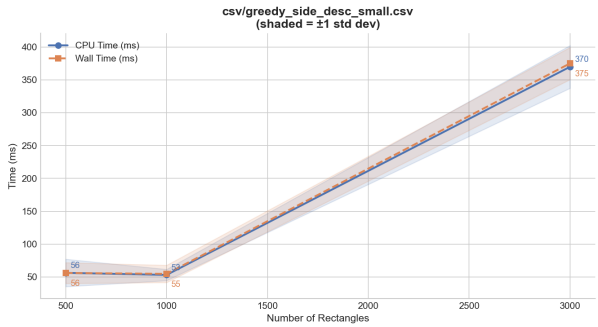
6 CPU vs Wall Time

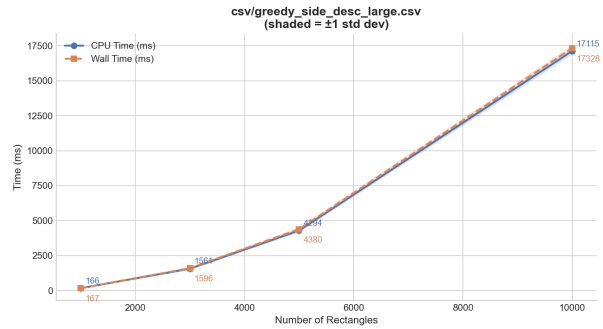
Greedy – Area Descending



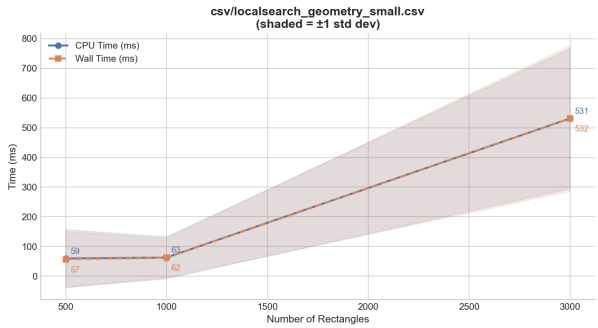


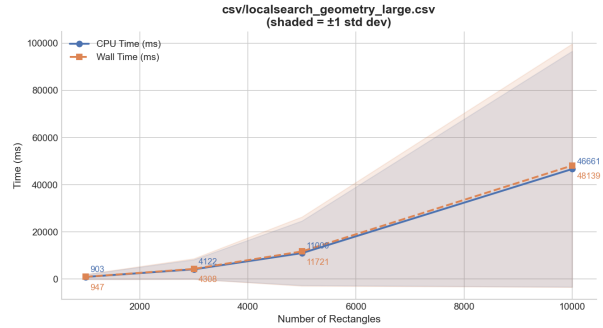
Greedy – Side Length Descending



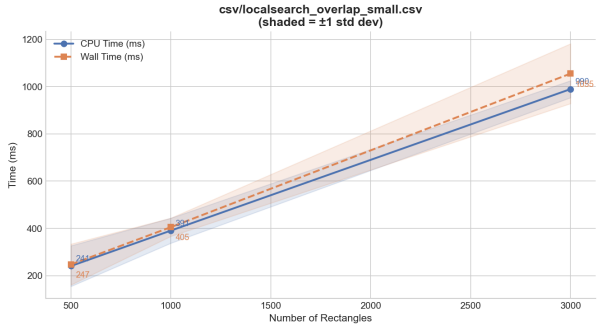


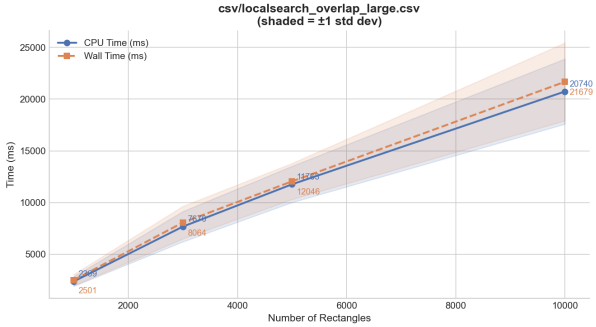
Local Search – Geometry



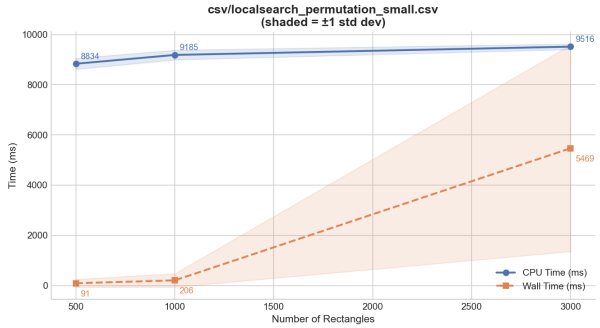


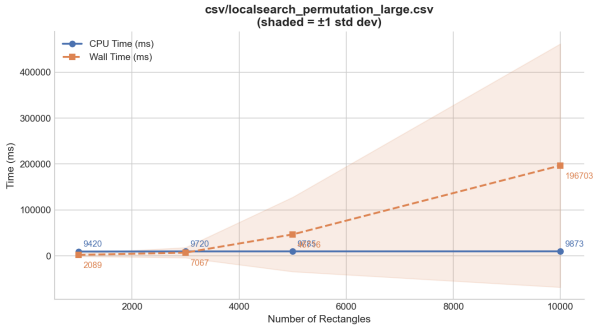
Local Search – Overlap





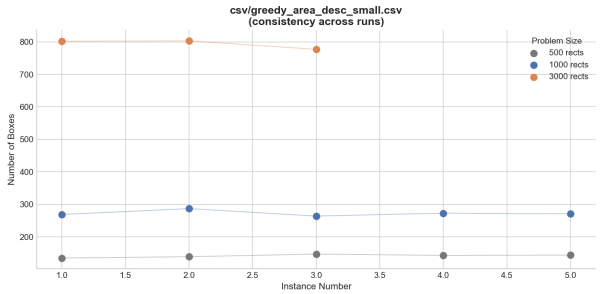
Local Search – Permutation

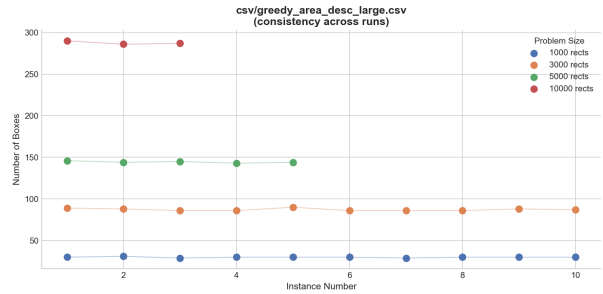




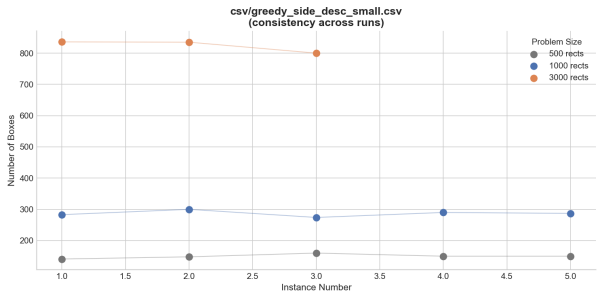
7 Boxes Per Instance

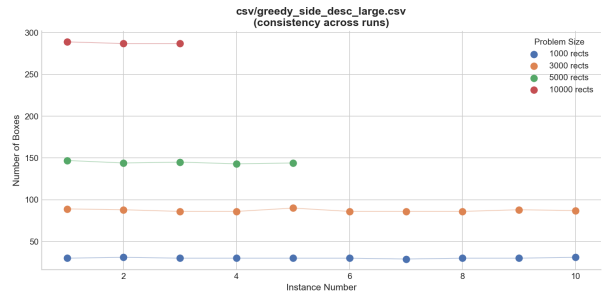
Greedy – Area Descending



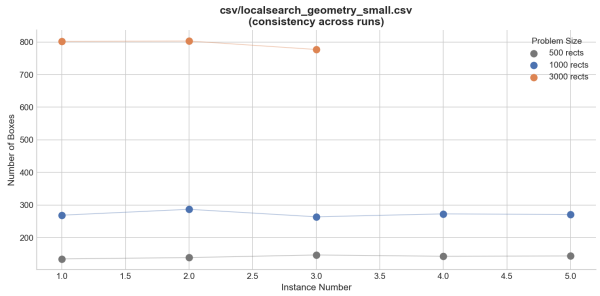


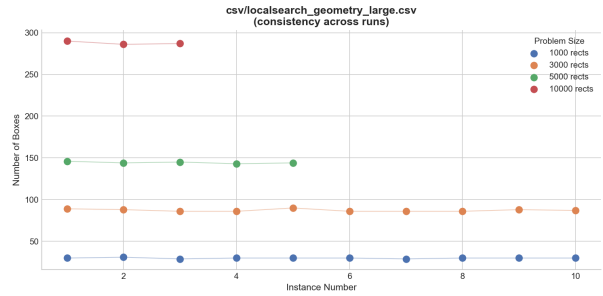
Greedy – Side Length Descending



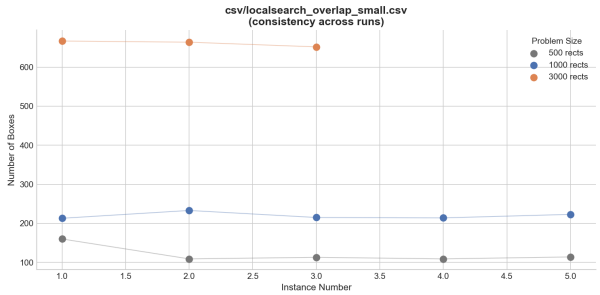


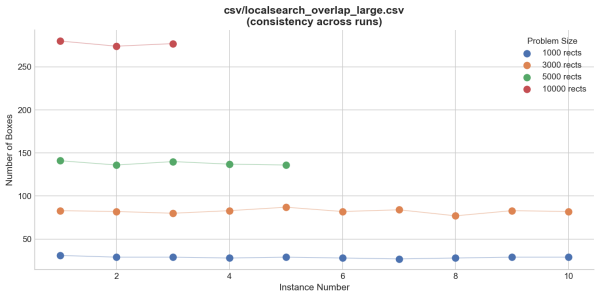
Local Search – Geometry



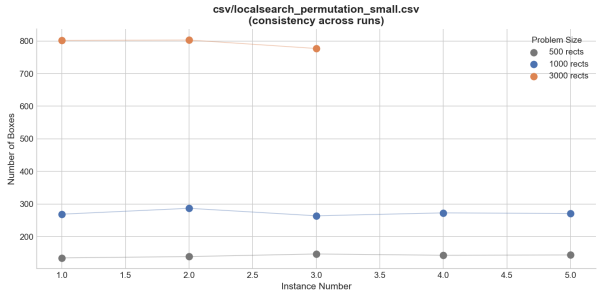


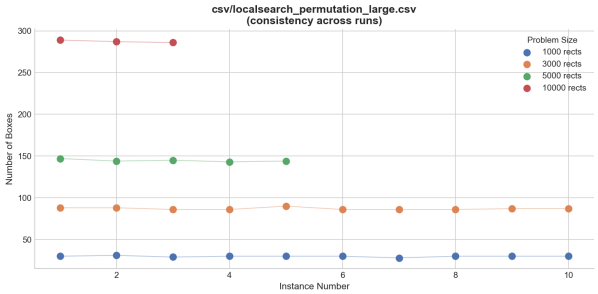
Local Search – Overlap





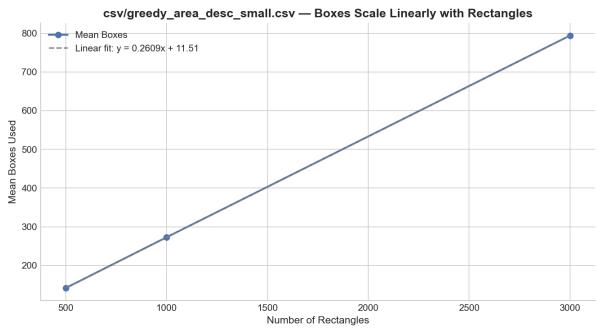
Local Search – Permutation

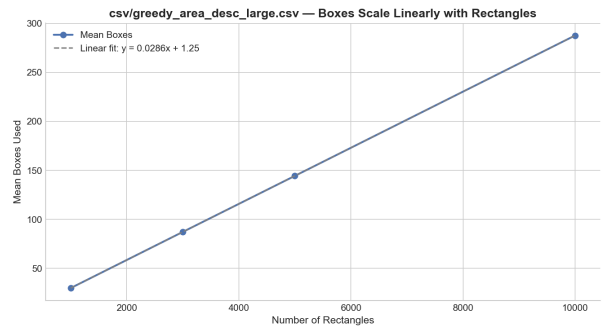




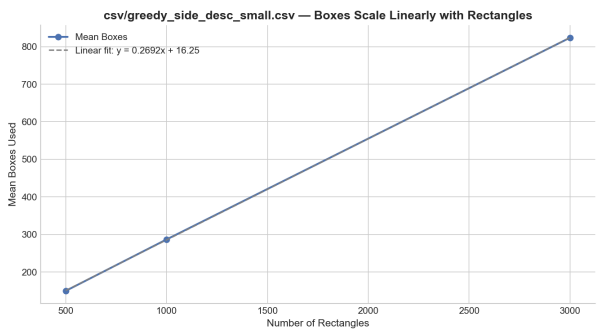
8 Scaling Behaviour

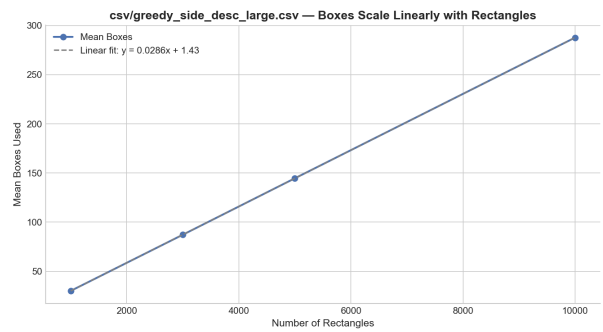
Greedy – Area Descending



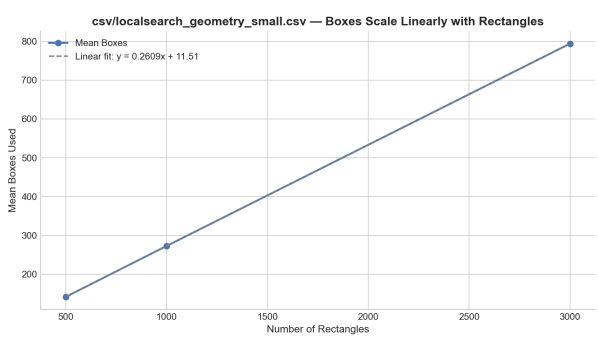


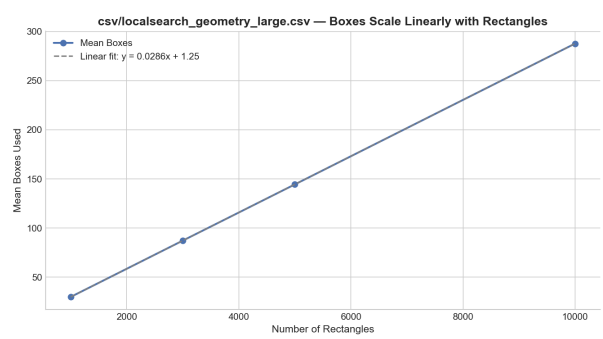
Greedy – Side Length Descending



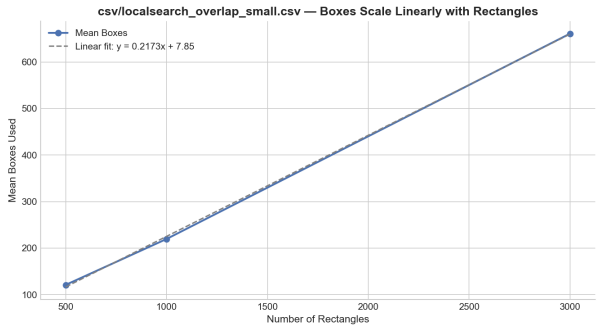


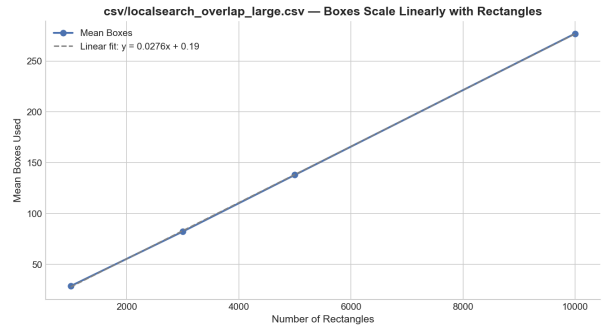
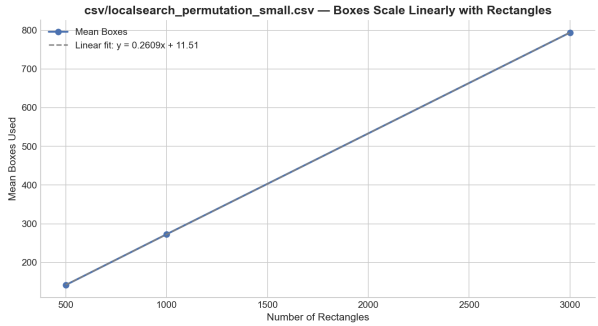
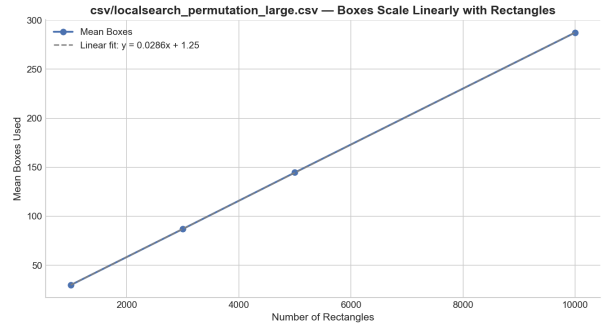
Local Search – Geometry





Local Search – Overlap





Local Search – Permutation